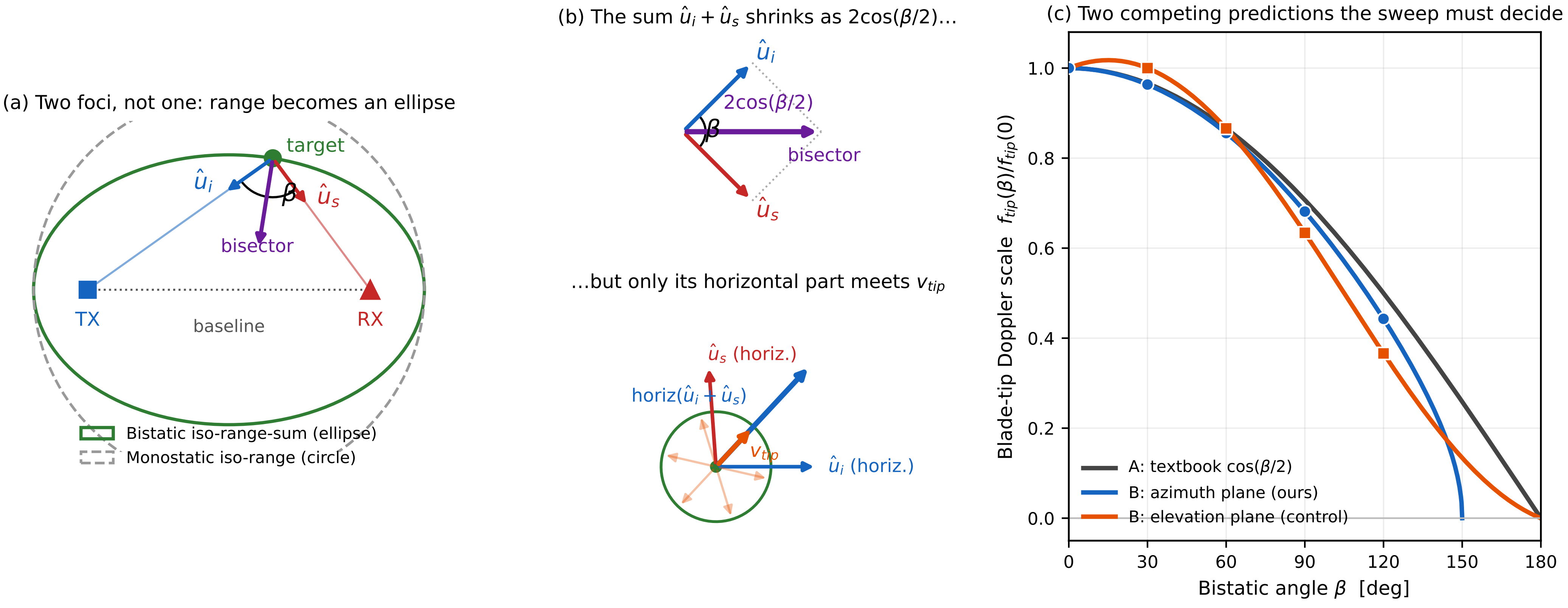

Passive bistatic geometry for the report-7 target and look direction. (a) The transmitter and the receiver are the two foci of the iso-range-sum ellipse, so a monostatic range circle becomes an ellipse and the target sees a bistatic angle beta with a bisector halfway between the two legs. (b) Doppler is set by the vector sum of the two unit directions, which has length 2cos(beta/2) along the bisector. The blade-tip velocity of a hovering rotorcraft lies in the horizontal rotor plane and sweeps every azimuth once per revolution, so the tip Doppler is fixed by the HORIZONTAL part of that sum, not by its length. (c) The two predictions therefore differ whenever the bisector tilts out of the rotor plane: A is the textbook shorthand, B is the exact projection for the geometry actually swept here. Panel (b) is drawn at beta = 90 deg in the azimuth plane; the marker positions in (c) are read from the sweep ledger, and the curves are the closed forms checked against it.